

# C. Vladimir Rodríguez-Caballero

Department of Statistics, ITAM

Río Hondo No.1, Col. Progreso Tizapán, Álvaro Obregón

CDMX 01080, Mexico

@ vladimir.rodriguez@itam.mx • W vlasmetrics7.github.io

vladimir.rodriguez@duke.edu · Scholar: [Link] · ORCID: 0000-0002-3601-7715 · Scopus: [Link]

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## Research Fields

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Time series econometrics (long memory, fractional integration, cointegration); high-dimensional factor models (multilevel dynamic factor models, FI-ML-DFM); scenario-based predictive densities (Growth-at-Risk, GiS/GaR); macroeconometrics and forecasting.

## Education

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Aarhus University / CREATES Aarhus, Denmark  
PhD in Econometrics 2013–2016  
School of Business and Social Science; Center for Research in Econometric Analysis of Time Series (CREATES).  
Dissertation: On factor analysis with long-range dependence. [Link]  
Advisor: Niels Haldrup.

Universidad de Guanajuato Guanajuato, Mexico  
M.Sc. in Economics (Econometrics) — Summa Cum Laude 2009–2011  
Dissertation: Ensayos sobre la Granger Causalidad. [Link]  
Advisor: Daniel Ventosa-Santaulària.

Facultad de Ciencias, UNAM Mexico City, Mexico  
B.Sc. in Actuarial Science 2001–2005  
Dissertation: Inferencia Bayesiana para la volatilidad en el modelo Black & Scholes. [Link]  
Advisors: Alejandro Villagrán Hernández and Ramsés H. Mena Chávez.

## Academic Appointments

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|------------------------------------------------------------------|---------------------|
| Duke University, Department of Economics                         | North Carolina, USA |
| Visiting Professor                                               | Aug 2025 – Jun 2026 |
| ITAM, Department of Statistics                                   | Mexico City, Mexico |
| Associate Professor (Tenured)                                    | Aug 2023 – present  |
| ITAM, Department of Statistics                                   | Mexico City, Mexico |
| Assistant Professor (Tenure Track)                               | Aug 2018 – Jul 2023 |
| Universidad Carlos III de Madrid, Department of Statistics       | Madrid, Spain       |
| Assistant Professor of Econometrics (Tenure Track)               | Sep 2017 – Jul 2018 |
| Universidad Carlos III de Madrid, Department of Statistics       | Madrid, Spain       |
| Postdoctoral Researcher                                          | Jan 2017 – Aug 2017 |
| Aarhus University / CREATES                                      | Aarhus, Denmark     |
| Research Assistant                                               | Sep 2016 – Nov 2016 |
| CREATES, Aarhus University / Danish National Research Foundation | Aarhus, Denmark     |
| PhD Fellow                                                       | Sep 2013 – Aug 2016 |

## Affiliations

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CREATES, Aarhus University  
Research Fellow

Aarhus, Denmark  
Nov 2016 – Dec 2022

## Professional Experience (Non-Academic)

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HSBC Holding  
Credit Risk Analyst / Credit Risk Manager  
Risk Management.

Mexico City, Mexico  
Feb 2006 – Jan 2009

Banco Fácil  
Credit Risk Manager  
Risk Management.

Mexico City, Mexico  
Mar 2007 – Jan 2009

Microfinanciera Creciendo  
Credit Risk Manager  
Risk.

Mexico City, Mexico  
Sep 2011 – Feb 2012

Santander Private Banking, Marketing Intelligence  
Subdirector, Statistical Models and Methodology

Mexico City, Mexico  
Apr 2012 – Mar 2013

## External Research Collaborations

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Interamerican Development Bank  
Pollution and education.

2025

Interamerican Development Bank  
Evaluación de la Fundación Camino a Casa.

2021–2022

Interamerican Development Bank  
Gender equality and female labor participation.

2020

Interamerican Development Bank  
Remittances payments through central banks.

2019

## Articles [Quartiles: JCR, Scimago, Scopus]

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- (24) Belloc, G., Garrón, I., Rodríguez-Caballero, C. V., & Ruiz, E. (2026). FARS: Factor Augmented Regression Scenarios in R. *Journal of Statistical Software*. [Forthcoming] [Q1, Q1, Q1]
- (23) Rodríguez-Caballero, C. V., & Ruiz, E. (2026). Temperature in the Iberian Peninsula: Trend, seasonality, and heterogeneity. *Annals of Applied Statistics*, 20(1), 173–196. [Q1, Q1, Q1] [Link]
- (22) Garrón, I., Rodríguez-Caballero, C. V., & Ruiz, E. (2026). International factors and inflation risks. *International Journal of Forecasting*, 42(3). [Q1, Q1, Q1] [Link]
- (21) Trillo, F. H., Rodríguez-Caballero, C. V., & Ventosa-Santaularia, D. (2025). Monopoly unveiled: Telecom breakups in the US and Mexico. *Economics Bulletin*, 45(4), 1989–1995. [~, Q3, Q3]
- (20) de Juan, A., Poncela, P., Rodríguez-Caballero, C. V., & Ruiz, E. (2025). Economic activity and climate change. *International Review of Environmental and Resource Economics*, 19(2). [Q3, Q3, Q3] [Link]
- (19) Gonzalez-Rivera, G., Rodríguez-Caballero, C. V., & Ruiz, E. (2024). Expecting the unexpected: Stressed scenarios for economic growth. *Journal of Applied Econometrics*, 39(5), 926–942. [Q1, Q1, Q1] [Link]
- (18) Caporin, M., Rodríguez-Caballero, C. V., & Ruiz, E. (2024). The factor structure of exchange rates volatility: Global and intermittent factors. *Empirical Economics*. [Q2, Q2, Q2] [Link]

- (17) López-Marmolejo, A., & Rodríguez-Caballero, C. V. (2023). Assessing the effect of gender-related legal reforms on female labour participation and GDP per capita in the Central American region. *Regional Statistics*, 13(3). [∼, Q2, Q2] [Link]
- (16) Ergemen, Y. E., & Rodríguez-Caballero, C. V. (2023). Estimation of a dynamic multi-level factor model with possible long-range dependence. *International Journal of Forecasting*, 39(1), 405–430. [Q1, Q1, Q1] [Link]
- (15) Prados de la Escosura, L., & Rodríguez-Caballero, C. V. (2022). War, pandemics, and modern economic growth in Europe. *Explorations in Economic History*, 86, 101467. [Q1, Q1, Q1] [Link]
- (14) Rodríguez-Caballero, C. V., & Villanueva-Domínguez, M. (2022). Predicting cryptocurrency crash dates. *Empirical Economics*, 63, 2855–2873. [Q2, Q2, Q1] [Link]
- (13) Rodríguez-Caballero, C. V., & Vera-Valdés, J. E. (2021). Air pollution and mobility: What carries COVID-19? *Econometrics*, 9, 37. [Q2, Q2, Q2] [Link]
- (12) López-Marmolejo, A., Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2021). Remittances at record highs in Latin America: Time to revisit the Dutch disease. *Economics Bulletin*, 41(3), 2133–2146. [∼, Q3, Q3] [Link]
- (11) Vera-Valdés, J. E., & Rodríguez-Caballero, C. V. (2021). Air pollution and mobility in the Mexico City metropolitan area in times of COVID-19. *Atmósfera*, 36(2), 343–354. [Q4, Q3, Q3] [Link]
- (10) Cataño, D. H., Rodríguez-Caballero, C. V., Chiann, C., & Peña, D. (2021). Wavelet estimation for factor models with time-varying loadings. *International Journal of Wavelets, Multiresolution and Information Processing*, 20(1), 2150033. [Q3, Q3, Q3] [Link]
- (9) Rodríguez-Caballero, C. V. (2021). Energy consumption and GDP: A panel data analysis with multi-level cross-sectional dependence. *Econometrics and Statistics*, 23, 128–146. [Q2, Q2, Q2] [Link]
- (8) Rodríguez-Caballero, C. V., & Vera-Valdés, J. E. (2020). Long-lasting economic effects of pandemics: Evidence on growth and unemployment. *Econometrics*, 8(3), 37. [Q2, Q2, Q2] [Link]
- (7) Rodríguez-Caballero, C. V., & Caporin, M. (2019). A multilevel factor approach for the analysis of CDS commonality and risk contribution. *Journal of International Financial Markets, Institutions and Money*, 63, 101144. [Q1, Q1, Q1] [Link]
- (6) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2017). Energy-growth long-term relationship under structural breaks: Evidence from Canada, 17 Latin American economies and the USA. *Energy Economics*, 61, 121–134. [Q1, Q1, Q1] [Link]
- (5) Ergemen, Y. E., Haldrup, N., & Rodríguez-Caballero, C. V. (2016). Common long-range dependence in a panel of hourly Nord Pool electricity prices and loads. *Energy Economics*, 60, 79–96. [Q1, Q1, Q1] [Link]
- (4) Rodríguez-Caballero, C. V., & Knapik, O. (2014). Bayesian log-periodic model for financial crashes. *The European Physical Journal B*, 87(10), 1–14. [Q2, Q3, Q3] [Link]
- (3) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2014). Granger causality and unit roots. *Journal of Statistical and Econometric Methods*, 3(1), 97–114. [Link]
- (2) Ventosa-Santaulària, D., & Rodríguez-Caballero, C. V. (2013). Polynomial regressions and non-sense inference. *Econometrics*, 1(3), 236–248. [Link]

- (1) Espin-García, O., & Rodríguez-Caballero, C. V. (2013). Metodología para un scoring de clientes sin referencias crediticias. Cuadernos de Economía, 32(59), 137–162. [Q3, Q4, Q3] [Link]

## Books

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- (1) Rodríguez-Caballero, C. V., & Espin García, O. (2012). Matemáticas Financieras II. Editorial GAFRA, Puebla, México. [Link]

## Book Chapters

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- (4) Rodríguez-Caballero, C. V., López Marmolejo, A., & Mendez Rolón, E. (2023). Efectos del Modelo de Atención de la Fundación Camino a Casa. In Modelo de atención para recuperar y empoderar vidas: quince años uniendo esfuerzos. Inter-American Development Bank. [Link]
- (3) Rodríguez-Caballero, C. V. (2023). Approximation of hate detection processes in Spanish and other non-Anglo-Saxon languages. In News Media and Hate Speech Promotion in Mediterranean Countries (pp. 65–80). IGI Global. [Link]
- (2) Rodríguez-Caballero, C. V. (s.f.). Granger causalidad espuria en la relación de cartera total y vencida de créditos. In Volumen IV: Mercados, modelos financieros y entorno económico. UAM.
- (1) Rodríguez-Caballero, C. V. (s.f.). La inferencia bayesiana en la administración de riesgos. In Volumen II: Mercados, modelos financieros y entorno económico, p. 181. UAM.

## Technical Reports

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- (2) Rodríguez-Caballero, C. V., López, A., & Ventosa-Santaulària, D. (2019). Remittances payments through central banks: An application to the Central American countries exchange rates. IDB Technical Notes, IDB-TN-1864. [Link]
- (1) Rodríguez-Caballero, C. V., & López, A. (2020). Assessing the effect of gender equality before the law on female labor participation and GDP per capita in Central America, Panama, and the Dominican Republic. IDB Technical Notes, IDB-TN-2128. [Link]

## Pop-Science Articles

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- (1) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. (2025). El impacto indeleble de la regresión. Miscelánea Matemática, 81.

## Software Development

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FARS: Factor-Augmented Regression Scenarios (2025). R package for modelling and forecasting economic scenarios based on multi-level dynamic factor models. Available on CRAN: [Link]

## Working Papers and Work in Progress

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- (3) Rodríguez-Caballero, C. V., Bellocca, G. P., Garrón, I., & Ruiz, E. The empirical distribution of sequential LS factors in multi-level dynamic factor models. R&R, Econometrics and Statistics.
- (2) Rodríguez-Caballero, C. V., Pérez Castellano, J. L., & López Marmolejo, A. Environmental inequality and educational outcomes: Evidence from air pollution in Mexico. Manuscript in preparation.
- (1) Rodríguez-Caballero, C. V., & Ventosa-Santaulària, D. Time Series Econometrics. [Book in preparation]

## Participation in Research Projects

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1. Incertidumbre en modelos de factores dinámicos: Un índice de riesgo climático. Ministerio de Economía y Competitividad, PID2022-139614NB-C22, 2023–2027.
2. Climate change and economic challenges for the Spanish society. La Caixa Foundation, 2020–2021.
3. Indicadores económicos: Predicción con incertidumbre e inestabilidad. Ministerio de Economía y Competitividad, ECO2015-70331-C2-2-R.

## Awards and Grants

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Mexican National System of Researchers (SNI) — Secihti<br>SNI Level II                                                                                                            | 2024 – present |
| Mexican National System of Researchers (SNI) — Secihti<br>SNI Level I                                                                                                             | 2018 – 2023    |
| ACRI and the Italian Econometric Association<br>Young Investigator Training Program (YITP) Prize                                                                                  | 2016           |
| International Association for Applied Econometrics<br>Travel Grant (competitive basis), 3rd IAAE Conference                                                                       | 2016           |
| CREATES, Aarhus University / Danish National Research Foundation<br>Full PhD Fellowship                                                                                           | 2013           |
| Universidad de Guanajuato<br>Distinction: Best Academic Record, MSc 2009–2011                                                                                                     | 2011           |
| CONACYT<br>Scholarship for research stay abroad                                                                                                                                   | 2011           |
| Universidad de Guanajuato<br>Recognition of Academic Merit (2010 & 2011)                                                                                                          | 2011           |
| CONACYT<br>CONACYT Graduate Scholarship (Master's)                                                                                                                                | 2009–2011      |
| Premio Nacional Francisco Aranda, Mexican Statistical Association<br>National Statistical Prize, 3rd place<br>Best bachelor dissertation in statistics, academic years 2005–2006. | 2005           |
| UNAM – Telmex Foundation<br>Excellence Scholarship                                                                                                                                | 2003–2005      |

## Teaching

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ITAM: Undergraduate: Econometrics I, Regression Analysis, Time Series Econometrics, Introduction to Statistics.  
Graduate (MSc): Financial Econometrics (Volatility), Time Series Econometrics, Advanced Econometrics (Asymptotic Theory), Climate Econometrics.

UC3M: Undergraduate: Statistical Methods for Social Sciences (Applied Time Series).  
Graduate (MSc/PhD): Causal Inference.

Aarhus University: Quantitative and Limited Dependent Variables; Mathematical Economics II. (Teaching assistant)

CIDE (2013): MSc in Economics: Introduction to Multiple Time Series Analysis.

## Supervised Students

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BSc: A. Jørgensen & M. Sproegel (Aarhus, 2015); G. Santa Rosa García (2016); M. Cortés (2 theses, 2019); M. Villanueva (2 theses, 2020); P. Peralta (2022); R. Gómez (2023); P. Gracia (2023); P. Galeana (2023); L. Palacios (2024); X. Iñiguez (2024); A. Natividad (2024); G. Salazar (2025); A. L. García (2025); F. Pérez (2026).

MSc: M. Cañizares (2018); H. Frech (2018); M. Cortés (2022); R. Garduño (2023); P. Gracia (2026).

## Research Stays

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|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Universidad Carlos III de Madrid<br>Department of Statistics                                                               | Madrid, Spain<br>Summer 2024          |
| Universidad Carlos III de Madrid<br>Department of Statistics                                                               | Madrid, Spain<br>Summer 2022          |
| Universidad Carlos III de Madrid<br>Department of Social Sciences                                                          | Madrid, Spain<br>Summer 2019          |
| Department of Statistical Sciences, University of Padova<br>Visiting PhD Scholar, ACRI YITP<br>Host: Massimiliano Caporin. | Padova, Italy<br>Spring 2016          |
| Department of Economics, Universidad Carlos III de Madrid<br>Visiting PhD Scholar<br>Host: Carlos Velasco.                 | Madrid, Spain<br>Spring 2016          |
| CIDE<br>Research visit                                                                                                     | Mexico City, Mexico<br>May – Jun 2013 |
| Institute of Mathematics, Cracow University of Technology<br>Research visit                                                | Cracow, Poland<br>Jan – Jul 2011      |
| Université Paris I (Panthéon-Sorbonne), SAMM Laboratoire<br>Research visit                                                 | Paris, France<br>Jun 2011             |

## Seminars and Conferences (from 2015)

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2025: Banco de Uruguay (Mar); Bank of England (Apr); Duke University (Oct); CIMAT Aniversario (Nov).

2024: 44th International Symposium on Forecasting, Dijon, France (Jul); 34th Midwest Econometrics Group Conference, Lexington KY, USA (Nov).

2023: Coloquio Mexicano de Economía Matemática y Econometría, Guanajuato (Sep); EcoSta 2023, Tokyo (Aug); Seminario ITAM; Seminario CIDE; Coloquio AME; Seminario CIMAT.

2022: Annual Conference IAAE, London; Arkansas State University–CQ; Regional Cooperation and Green Economy Conference, Liaoning University, China.

2021: Department Seminar, Örebro University, Sweden; 14th Workshop on Nonstationary Systems, Grodek, Poland; 26th International Panel Data Conference; XXXIV Foro Nacional de Estadística; CLATSE 2021; Universidad Veracruzana.

2020: International Symposium on Forecasting.

2019: Latin American Meeting of the Econometric Society, Puebla; Annual Conference IAAE, Cyprus; International Panel Data Conference, Vilnius.

2018: VIII Workshop in Time Series Econometrics, Zaragoza; Workshop on Econometrics and Data Science, Monterrey; Long Memory Conference, Aalborg; Semana de Matemáticas, ITAM; Department Seminar, Banco de México; Seminario Aleatorio, ITAM.

2017: Department of Statistics, UC3M, Madrid; 25th Annual Symposium SNDE, Paris; 40th Interna-

tional Panel Data Conference, Thessaloniki; 70th ESEM, Lisbon; Department Seminar, Universidad Pública de Navarra.

2016: LNPP Seminar, CIDE, Mexico City; Department of Statistical Sciences, University of Padova; CREATES Anniversary Meeting, Sandbjerg; Annual Conference IAAE, Milan; Workshop in Time Series Econometrics, Zaragoza; PhD Workshop, Madrid.

2015: CREATES Lunch Seminar (Nov 2015 & Feb 2016), Aarhus; Annual DGPE Workshop, Sandbjerg.

## Referee and Editorial Activities

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Associate Editor: Latin American Economic Review (LAER), from October 2023.

Referee for: Trimestre Económico; Lecturas de Economía; Revista de Métodos Cuantitativos para la Economía y la Empresa; International Journal of Forecasting; The American Statistician; Latin American Economic Review; Journal of the Operational Research Society; Energy Economics; Empirical Economics; Journal of Nonparametric Statistics; Resources Policy; Energy Journal; Nature; Quantitative Finance; Journal of Econometrics; Regional Statistics; International Journal of Environmental Research and Public Health; Energies; Applied Economics Letters; Environment and Development Economics; Journal of Economic Structures; Atmosphere; Finance Research Letters; Communications in Statistics; Econometrics and Statistics; Financial Innovation; Communications in Statistics — Simulation and Computation; Revista de Matemática Teoría y Aplicaciones; Computational Statistics and Data Analysis.

## Organization of Academic Events

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- Department Seminar at ITAM. Principal organizer.
- Coloquio Virtual AME, 2020. Principal organizer.
- Mathematical and Statistical Methods for Actuarial Sciences and Finance (MAF 2018), Madrid. Member of the Scientific Committee.

## Service in Selection Committees and Academic Societies

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- Member, Comisión Dictaminadora 2, Área VI: Ciencias Sociales. Secihti.
- Premio Francisco Aranda Ordaz, AME, 2020–2021. Principal organizer.
- Academic evaluator for Secihti calls (PhD studies, postdoctoral stays, scientific projects).
- Member of multiple selection committees, Fulbright–García Robles program, COMEXUS.

## References

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Available upon request.